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Green biolubricant infused slippery surfaces to combat marine biofouling



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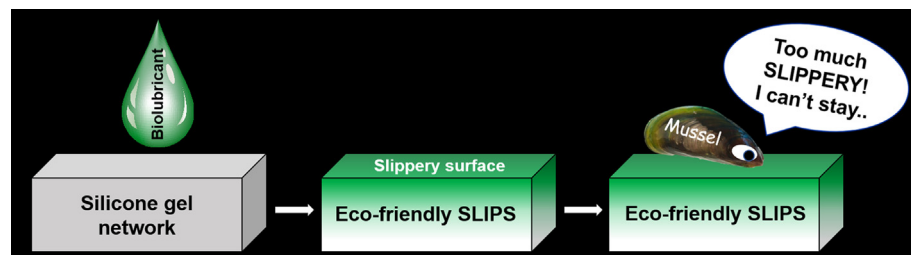
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GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: Marine biofouling is a global, longstanding problem for maritime industries and coastal areas arising from the attachment of fouling organisms onto solid immersed surfaces. Slippery Liquid Infused Porous Surfaces (SLIPS) have recently shown promising capacity to combat marine biofouling. In most SLIPS coatings, the lubricant is a silicone/fluorinated-based synthetic component that may not be fully compatible with the marine life. We hypothesized that eco-friendly biolubricants could be used to replace synthetic lubricants in SLIPS for marine anti-fouling.

Experiments: We developed SLIPS coatings using oleic acid (OA) and methyl oleate (MO) as infusing phases. The infusion efficiency was verified with confocal microscopy, surface spectroscopy, wetting efficiency, and nanocontact mechanics. Using green mussels as a model organism, we tested the anti-fouling performance of the biolubricant infused SLIPS and verified its non-cytotoxicity against fish gill cells.

Findings: We find that UV-treated PDMS infused with MO gives the most uniform infused film, in agreement with the lowest interfacial energy among all surface/biolubricants produced. These surfaces exhibit efficient anti-fouling properties, as defined by the lowest number of mussel adhesive threads attached to the surface as well as by the smallest surface/thread adhesion strength. We find a direct correlation between anti-fouling performance and the substrate/biolubricant interfacial energy.

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1. Introduction

The colonization of marine organisms onto man-made surfaces, called marine biofouling, such as ship hulls [1], has major economical as well as environmental consequences for maritime-related fields. It has been estimated that the increase in hydrodynamic drag associated with vessel hull fouling results in fuel consumption increase by as much as 41% [2], or an extra 300 million tons of fuels [3] producing an estimated 20 million tons/year in additional greenhouse gases. Worldwide, the cost associated to the shipping industry due to biofouling has been estimated to be around USD 30 billion per year [1]. In addition, the attachment of marine organisms deteriorates submerged facilities (costal power plants, water pipelines, fisheries, etc.) due to corrosion induced both by metabolites produced by the fouling organisms as well as the degradation of anti-corrosion coatings, resulting in exposure of the underlying surface to the aggressive seawater environment [4,5]. The economic loss caused by biocorrosion is also extremely onerous. Therefore, inhibiting biofouling is one of the most important challenges faced by any solid surfaces immersed in the marine environment. Historically, antifouling coatings containing biocides such as tributyltin (TBT) were very efficient in deterring biofouling, especially in marine engineering applications. However, such coatings resulted in unacceptable toxicity towards marine ecology [6], which notably led to the ban of TBT-containing coatings by the International Maritime Organization [7].

Nature affords useful inspiration for a variety of technological challenges, including for biofouling resistance. Lotus leaf can repel liquid droplets by the so-called superhydrophobic effect, leading to high resistance against surface wettability, a characteristic that researchers have replicated to produce superhydrophobic materials for water repellency applications [8]. Superhydrophobic materials have thus been explored as potential candidates to avoid biofouling with some success [9,10]. It is widely accepted that the anti-biofouling property of a superhydrophobic surface is due to the entrapped air cushion that reduces contact between the adhesive appendage of the organism and the solid substrate [10]. However, the prolonged immersion of superhydrophobic surfaces in water may lead to the loss of the entrapped air layer, and eventually fouling organisms can settle on such surfaces in the field. Furthermore, because superhydrophobic surfaces are intrinsically rough, they may also actually encourage adhesion of organisms over long-term exposure [10].

Recently, the concept of “Slippery Liquid-Infused Porous Surfaces” (SLIPS) has emerged as a new class of repellent coatings with highly promising antifouling capacity [11,12], including for marine biofouling [13]. The technology consists in infusing a porous substrate, either a micro- or nano-structured surface or a polymeric gel, with a low-surface energy fluid (the “synthetic lubricant”), which remains entrapped within the porous substrate owing to the strong chemical affinity between substrate and lubricant. In a recent study, we notably established that polydimethylsiloxane (PDMS)-based SLIPS coatings exhibit exceptional anti-fouling capacity against aquatic mussels (one of the most aggressive macrofouling organism) both in the lab and in the field, by effectively deterring mussel settlement in the first place, but also by greatly reducing the adhesion strength of the mussel attachment threads after settlement, which is beneficial for their easy removal. However, commercially available synthetic oils/lubricants are usually fluorinated and their slow leakage over time is a matter of concerns from an environmental perspective [14]. Alternative eco-friendly biolubricants would thus expand the potential of SLIPS for marine applications.

Non-toxic, eco-friendly fatty acid based biolubricants have been considered to prepare bacterial biofilm resistant materials [15,16]. For example, it was observed that a mixture of palmitic, stearic,

oleic and linoleic acids was able to inhibit biofilm formation, disturbing the microbial quorum sensing, thus indicating that fatty acids inhibit biofilm formation and, in turn, inhibit larval settlement [17]. In another recent study, Awad et al. [18] have developed stainless steel-based SLIPS infused with vegetable edible oils and demonstrated promising antibiofilm activity of the oil-infused surface. Furthermore, it was established by Kang et al. [19] that oleamide, an amidated version of oleic acid, is a major component of the periostracum of marine mussels (the organic coating covering their shells) and that it played an important role in minimizing fouling of the shell itself, notably against algal spore settlement.

Based on these studies, we posited that the unsaturated fatty acid oleic acid (OA) and its ester derivative methyl oleate (MO) could constitute potential antifouling bio-based lubricants as substitutes to synthetic fluorinated oils used in the majority of SLIPS coatings to date [20]. Indeed, unsaturated fatty acids and their ester derivatives are abundantly available in nature and are also considered as renewable and alternative lubricants [21]. To obtain an oleic acid or methyl oleate lubricant-saturated water-repellent surface, an elastomeric PDMS gel was used, similar to our previous work [13]. After lubricant infusion, the PDMS gel will swell, trapping the lubricant oil in its pores by capillarity and creating a slippery lubricant layer. In our previous work, we showed that synthetic lubricant-infused PDMS are able to efficiently prevent mussel fouling based on purely physical phenomena, namely shielding of the underlying solid substrate from mussels' detection and very low adhesion strength. Here, we hypothesize that a similar anti-fouling performance could be achieved with biolubricants, as long as the lubricants efficiently infuse the PDMS substrates, remain trapped within the porous network, and at the same time form a smooth liquid layer over the underlying gel substrate. Using confocal fluorescence imaging and Attenuated Total Reflection Fourier Transform InfraRed (ATR-FTIR) spectroscopy, we find that both biolubricants can infuse PDMS, although the lubricated layer appeared non-uniform. However, with a UV treatment of PDMS prior to infusion, more efficient and uniform infusion could be achieved, in particular for MO-infused PDMS owing to a surface tension closer to that of PDMS than OA, which is a critical parameter for stable lubricant infusion. Using contact nanomechanical experiments, permanent infusion of MO within the PDMS network was further substantiated: the biolubricant forms a capillary adhesive bridge upon disengagement of a spherical tip. In addition, well-swollen network (which is a direct consequence of infusion) results in a decreased in elastic modulus as inferred from Hertzian contact mechanic measurements. Finally, biofouling laboratory assays using the Asian green mussel *Perna viridis* (*P. viridis*) together with adhesive strength measurements of *P. viridis* adhesive threads on the surfaces indicate that our biolubricant-infused surface exhibits an anti-biofouling performance nearly equivalent to that of synthetic lubricated surfaces. Overall, our results show that fatty acids biolubricants represent a promising alternative to synthetic fluorinated lubricants for SLIPS applications, notably to combat marine biofouling.

2. Materials and methods

2.1. Preparation of biolubricant infused SLIPS

A Sylgard 184 elastomer kit (Dow Corning Corporation, USA) was used to prepare PDMS-coated glass slides. The base and curing agent were mixed in a 10:1 ratio for 2 min by using vortex (Biofrontier Technology, Singapore) and applied on a clear glass slide ($75 \times 25 \text{ mm}^2$) activated for 30 min in a UV/ozone chamber (Novascan, USA). One side of the glass slide was coated with PDMS solution using a spin-coater (Laurell Technologies Corporation,

USA) at 700 rpm for 30 s using, and the coated samples were dried in an oven (Memmert, Germany) at 60 °C for 4 h. The coated porous PDMS glass surfaces were infused in the presence of 10 $\mu\text{l}/\text{cm}^2$ of silicone oil (10 cSt Trimethylsiloxy terminated, Gelest) at room temperature to make SLIPS (Slippery Liquid-Infused Porous Surface) as described before [13]. Silicone oil infused PDMS sample was named as iPDMS.

Natural oils, OA (Sigma-Aldrich, USA; purity: 90%) and its ester derivative MO (TCI, Japan; purity: ~60%) were also used to prepare SLIPS-coated glass slide samples. PDMS and ultraviolet (UV)-treated PDMS (30 min UV ozone treatment) surfaces were infused in the presence of 20 $\mu\text{l}/\text{cm}^2$ OA and MO solutions to obtain four different biolubricants-infused SLIPS. After infusion, OA (oPDMS/oUV-PDMS) and MO (mPDMS/mUV-PDMS) infused samples were stored at 60 °C for 24 h. Excess lubricant was removed gravimetrically by tilting the surfaces at a near 90° angle. Subsequently, the samples were stored at room temperature for 24 h for further studies.

2.2. Confocal microscopy

To visualize the wetting and swollen states of the gels, we used confocal microscopy techniques in both the reflection and fluorescence modes (Supplementary Fig. S1a). We scanned the surface with monochromatic light of wavelength $\lambda_{\text{ex}} = 561$ nm using a galvanometric mirror and captured the reflected light through the pinhole of a confocal microscope (Olympus FV3000, Japan). At the same time, we captured the fluorescence signal (collected $\lambda_{\text{em}} = 565\text{--}650$ nm) from the lubricants (OA and MO) with added Oil Red O dye (Sigma Aldrich, USA) in the lubricant solution (1.0 mg/ml). The objective used had a magnification of $M = 10\times$ and a numerical aperture of $NA = 0.4$. The size of the pinhole was kept at about 2 Airy Units. This allowed us to achieve optical sectioning and collect light from a thickness of about $d_z = 15$ μm .

The results of these measurements and confocal cross-sections (corrected for the refractive index of the gel $n_{\text{gel}} = 1.4$) for two representative wetting states WS1 and WS2 in this study are shown in Supplementary Fig. S1b, c. The gel-glass and gel-air interfaces can be identified from the maxima in the reflected signals (averaged over a view of 1.3×1.3 mm, green curves in Supplementary Fig. S1b, c). The peak due to the glass-gel interface (A) is lower than the gel-air interface (B), because of the reduced refractive index contrast ($n_{\text{glass}} - n_{\text{gel}} = 0.1$ vs $n_{\text{gel}} - n_{\text{air}} = 0.4$). Although the intensity peak is broad with $d_z = 15$ μm , the maxima can be identified with 1 μm resolution, allowing us to measure thicknesses of the gels h before and after lubricant infusion with micrometer resolution.

From the fluorescence micrographs of different surfaces, we can identify two distinct wetting states. For wetting state 1 (WS1), the lubricants form discrete microdroplets over the PDMS gel. In contrast, for wetting state 2 (WS2), the lubricant forms a smooth, uniform overlayer. The intensity of the fluorescence signal (and hence the lubricant concentration) is constant inside the bulk of the gel. At the gel-air interface, there is a large fluorescence peak wherever there is a lubricant overlayer. Thus, the peak at B corresponds to the presence of lubricant microdroplets in WS1 and to a uniform lubricant film in WS2 (red full line in Supplementary Fig. S1b, c). For WS1, if we look at the fluorescence intensity signals only in the dry regions without microdroplets, we no longer find any peak at B (red dashed line in Supplementary Fig. S1b). Even for WS1, a thin nanolayer of oil may be present over the surface, which can be detected by depth-sensing nanoindentation as described in Section 2.9.

2.3. Contact angle (CA) measurements

CA were measured using a goniometer (OCA 15 Pro, DataPhysics, Germany) at ambient conditions. A 5.0 μl water droplet was

added to the surface (either plain PDMS or lubricant-infused) and the contact angle was extracted from the software using a Young-Laplace algorithm [22]. All measurements were repeated at least five times on different areas of the substrates and averaged. Contact angle hysteresis $\Delta\theta$ (that is, the difference between the advancing $[\theta_{\text{adv}}]$ and receding $[\theta_{\text{rec}}]$ contact angles of a moving droplet) was calculated by using the software SCA20. In measuring the contact angle hysteresis, the surface was tilted with respect to the horizontal plane until the liquid droplet started to slide along the surface. The droplet profiles were fitted into a spherical cap profile by SCA20 software in order to determine the advancing and receding angles, the sliding angle of the droplet (surface tilt base, TB, required for the liquid droplet motion), as well as the droplet volume.

CAs of oil droplets (OA and MO) on PDMS and UV-PDMS were measured using the same goniometer at ambient conditions. A 5.0 μl oil droplet was added to the surface (either plain PDMS or UV-PDMS) and the contact angle was extracted from the software using a Young-Laplace algorithm. All measurements were repeated at least five times on different areas of both substrates and averaged.

2.4. Surface energy and interfacial energy

Deionised water, diiodomethane (Sigma-Aldrich, USA; purity: 99%) and toluene (Sigma-Aldrich, USA; purity: 99.8%) have well-known surface tension values, and the relative contributions of both dispersive and polar components are also known [23]. These standard liquids were used to determine the surface energy (SE) of PDMS, iPDMS, oPDMS, mPDMS, UV-PDMS, oUV-PDMS and mUV-PDMS samples using the Owens, Wendt, Rabel and Kaelble (OWRK) method [24]. The OWRK method uses the geometric mean approach to combine the dispersion and polar components for the calculation of SE using equation (1):

$$\gamma = \gamma^d + \gamma^p \quad (1)$$

where γ is the total surface free energy and γ^d and γ^p are the dispersion and polar components of the surface energy, respectively. The interfacial energy (γ_{sl}) was calculated using the Girifalco and Good equation:

$$\gamma_{sl} = \gamma_s + \gamma_l - 2(\gamma_s^d \times \gamma_l^d)^{1/2} \quad (2)$$

where γ_s and γ_l are the surface free energies of substrate (PDMS or UV-PDMS) and lubricant (silicone oil/OA/MO), respectively and the superscript “d” denotes the dispersive component of the total interfacial energy [23].

2.5. Fourier transform infrared spectroscopy (FTIR)

FTIR spectra were collected using a FTIR spectrometer (Bruker Vertex V70, Germany) equipped with an MVP-PRO Attenuated Total Reflection (ATR) unit (diamond crystal, 45° angle of incidence) and a KBr beam splitter. All spectra were collected from the oven-dried samples (UV-PDMS, OA, MO, oUV-PDMS and mUV-PDMS) in the region of 400–4000 cm^{-1} with a 4 cm^{-1} resolution and averaged over 128 scans. The baseline correction of FTIR spectra was performed using concave rubber band correction method with OPUS software (Version 6.5, Bruker optics incorporation). SLIPS samples were carefully wiped off to remove excess oil from the surfaces before the FTIR measurements.

2.6. Mussel collection and preparation

Green mussels (*P. viridis*) were collected from an offshore mussel farm located near Lorong Halus Industrial Park, Singapore.

Adult mussels of 4–5 cm in length were selected and thoroughly cleaned using seawater. The mussels were immediately kept in shallow tanks of an artificial seawater aquarium system and equilibrated for three days at constant running seawater prior to the experiments. For the entirety of the test, the water was kept at an optimum temperature of 27–29 °C, pH 7.9–8.1 and a specific gravity between 1.022 and 1.023. In our checkerboard assays and adhesive strength testing, live mussels were scattered on the test surfaces.

2.7. Checkerboard mussel-choice assays

To assess the mussel's preference for substrates and identify the anti-fouling capacity of the coatings, three 6×4 checkerboards with randomly positioned surfaces (UV-PDMS, iPDMS, oUV-PDMS and mUV-PDMS) were prepared. Each board consisted of 6 slides of 4 surface types, with slides randomly distributed on each checker. The checkerboards were mounted on solid supports to minimize the exposure of the mussels to other surface choices. A total of 17 mussels were placed on each of the choice assay, immersed in artificial seawater and their behavior was observed throughout two weeks. Statistical analysis was obtained in terms of number of adhesive plaques planted on each surface.

2.8. Adhesive strength measurements

A customized micro-tensile machine was used to measure the pull-off force (plaque/substrate separation force) of individual mussel plaques as previously described [13]. The machine was equipped with a 100 g load cell (Futek, USA) connected to a computer through an external USB output kit for real time measurements of applied loads. Mussel byssal threads were cut with a clean sterile razor from the mussel proximal end and were kept hydrated in seawater prior to the measurements. The glass slides were positioned upside down in a custom-designed holder of the micro-tensile machine, and the free byssal threads were wrapped around a cylindrical holder at their distal proximity. All tested threads were longer than 1 cm since shorter threads were difficult to grip. The holder was slowly lifted and data were collected until the plaque detached from the substrate, corresponding to the maximum load at failure. All threads were pulled at an angle of $90^\circ \pm 5^\circ$ relative to the surface, and the plaque orientation relative to the thread remained identical for all threads. The pulled-off plaque traces were subsequently marked and their areas were measured using an optical microscope (Axio Scope.A1, Zeiss, Germany) equipped with an imaging software (AxioVision version 4.8.2, Zeiss). The adhesion strength on each test surface was measured by using the maximum applied load at failure normalized by the plaque's area. For all surfaces (UV-PDMS, oUV-PDMS, mUV-PDMS), adhesion strength tests were conducted on threads deposited on samples from at least five mussels and the mean value of all data was reported. We only took into account adhesive failure whereas cohesive failures (which occurred very rarely) were disregarded.

2.9. Contact forces at the nano-scale

Two types of experiments were explored using a Triboindenter TI-950 nanomechanical tester (Hysitron, USA) to assess the contact mechanic characteristics of the OA- and MO-infused UV-PDMS surfaces. For the infused surfaces, cono-spherical tips with nominal radii of 10 μm (for elastic modulus measurements) and 50 μm (for capillary adhesion force measurements) were employed, together with an extended displacement stage, allowing for maximum displacements of up to 500 μm . A 2D transducer with maximum force of 12 mN was used for the force measurements. To

detect the “jump-in/jump-off” effect and capillary bridging, experiments with the “air-indent” imaging mode were conducted as described before [13]. This “air-indent” imaging mode allows the recording of any force jump associated during the contact event, which was produced due to the “jump-in” effect. The extracted unloading curve was used to obtain the capillary adhesion force (maximum negative force measured during an unloading cycle), whereas the elastic modulus of the infused samples was obtained by fitting the loading portion of loading/unloading curves using the Hertzian equation [25].

2.10. Cell culture and cytotoxicity

The RTgill-W1 cell line from rainbow trout gill filaments was chosen for cell culture studies as a proxy to predict fish toxicity [26]. The cell line was purchased from ATCC® (CRL-2523™) and cells from passage numbers between 14 and 17 were used for testing. Fish gill cells were routinely cultured, expanded and maintained at 19 °C without CO₂ supplementation in 25 cm² and 75 cm² cell culture flasks using Leibovitz's L-15 media (ATCC® 30-2008) supplemented with 10% fetal bovine serum (FBS) (Gibco, USA).

2.11. Preparation of leachate

Glass slides coated with oUV-PDMS, mUV-PDMS and UV-PDMS were soaked in 1.0 ml ultrapure water (PURELAB flex 3, Veolia, UK) per square centimetre of coated surface for one week at room temperature, allowing any leachate to dissolve into the water. After soaking, the water containing leachate was freeze-dried and then stored at 4 °C before use. For the toxicity assay, 15 ml of complete L-15 medium was added to the tubes containing the freeze-dried leachate and put into a 37 °C incubator for 1 h to allow the leachate to dissolve into the medium. The assay media were prepared for test surfaces (oUV-PDMS and mUV-PDMS), cell death negative control (UV-PDMS) and cell death positive control (1% Triton-X). The assay medium was then passed through a 0.2 μm syringe filter before use in the toxicity assay.

2.12. Cell culture preparation and toxicity assay

Before adding to the assay media, confluent RTgill-W1 cells were washed with $1 \times \text{PBS}$ (Gibco, USA) twice, trypsinized (0.25% trypsin, Gibco, USA) at 37 °C for 5 min, pelleted at 200g for 5 min, and resuspended in 1.0 ml L-15 medium. The cells were counted using Countess™ II Automated Cell Counter (Invitrogen, USA) and diluted to a density of 300,000 cells/ml before being seeded into 24 well multidishes (Nunc, USA). The cells were allowed to attach for 24 h before L-15 medium was replaced by assay medium. After 24, 48 and 72 h, the assay medium was transferred to a clean 24 well multidish along with the Lactate Dehydrogenase (LDH) detection kit while the cells were washed gently with $1 \times \text{PBS}$ before testing with alamarBlue™. For the alamarBlue™ test, 526 μl of alamarBlue™ was added to 10 ml of complete L-15 medium before 500 μl of the solution was added to the assay wells. The cells were incubated for 1 h, protected from light, before the fluorescence was measured from each well at excitation/emission wavelengths of 530/590 nm, respectively, in a microplate reader (Infinite® 200 PRO, Tecan, Switzerland). For LDH detection, the transferred cell culture supernatant was centrifuged at 250g for 10 min. From each well, 100 μl of supernatant was transferred each to 3 wells of a 96 well multidish (Nunc, USA) before the test was performed as per the manufacturer's instructions. The absorbance at 490 nm was measured with the microplate reader.

3. Results

3.1. Surface characterization

Untreated PDMS surface showed a mean water CA of 109° (Supplementary Fig. S2a) indicative of its hydrophobic character under static conditions ($TB = 0^\circ$). Water droplets did not roll on the PDMS surface even after 1 min at 30° tilting angle (Fig. S2b) confirming that the unfused surface has no slippery character although it is hydrophobic in nature (Supplementary Movie S1).

We first compared the wetting states of untreated PDMS infused with OA and MO as biolubricants (oPDMS and mPDMS) and their counterparts that had been pre-treated with UV light (oUV-PDMS and mUV-PDMS). To observe the efficiency of biolubricant infusion, the surfaces were observed by confocal microscopy in both the fluorescence and reflection modes. Briefly, we shone the surface with a focused beam of monochromatic laser light of wavelength $\lambda = 561$ nm and captured both the reflected and fluorescence signals through the pinhole of a confocal microscope. The biolubricants were stained with the fluorescent dye Oil Red O (emission wavelengths λ_{em} between 565 and 650 nm), allowing us to visualize the spatial location of the gel. At the same time, we are able to accurately measure the coating thickness with micrometer accuracy from the light reflected off the glass-gel and gel-air interfaces (Fig. 1); see Materials and Methods for details.

Experimentally, we found that for oPDMS and mPDMS, the lubricant infusion was only partial (Supplementary Fig. S3a–b), with relatively small swelling ratio $h_f/h_i = 1.0$ – 1.3 even at elevated temperature of 60°C , where h_f and h_i are the initial and final film thicknesses. Moreover, the lubricant did not form a smooth, uniform overlayer, but instead dewetted into small, discrete microdroplets. We then hypothesized that enhanced infusion could be achieved by increasing the hydrophilicity of the PDMS substrate, which would increase its surface energy and in turn reduce the interfacial energy of PDMS with the biolubricants. A well-established approach to increase the hydrophilic properties of PDMS surfaces is to oxidize the polymer surface with plasma or UV irradiation [27], owing to the formation of a smooth oxidized surface layer of SiO_x [27,28]. The hydrophilicity of PDMS after such

treatments can be retained for an extended period of time in the presence of polar molecules (water or even ethanol) [29].

PDMS substrates were thus irradiated with UV light for 30 min prior to infusion. For OA, we found little to no change in the wetting behavior and swelling ratio of the gels (Fig. 1b, d). However, for MO, the lubricant infusion was greatly enhanced. Instead of forming discrete patches, the lubricant formed a smooth uniform overlayer (Fig. 1c) and the swelling ratio increased from 1.3 ± 0.1 for mPDMS to 1.9 ± 0.4 for mUV-PDMS (Fig. 1d).

The change in wetting properties are also reflected in the contact angle values on the surface before and after UV treatment. After 30 min of UV treatment, the PDMS surface showed a mean water CA of 101° ($<8^\circ$ lower than native PDMS) and higher surface wettability (Fig. 2a), indicating a slightly lower hydrophobic character compared to the untreated PDMS surface in static conditions, which in turn resulted *ca.* 22% increment of SE value (24.5 mN/m) compared to the native PDMS (19.13 mN/m) (Table 1). In terms of wetting behavior, UV-PDMS surface infused with OA lowered the mean CA to 77° (Fig. 2b), which can be attributed to the presence of the carboxylate group of OA. The mean CA was higher (84°) for mUV-PDMS, likely due to the substitution of the ionic carboxylate group by the less ionic ester moiety ($-\text{C}-\text{COO}-\text{C}-$) (Fig. 2c). The infusion characteristics were also evaluated by measuring the CAs of the biolubricants onto the substrates. On PDMS, the CAs of OA and MO were 57° and 29° , respectively (Supplementary Fig. 4a, c). After UV irradiation, these values decreased to 42° and 7° , respectively. (Supplementary Fig. 4b, d). These results corroborate the confocal microscopy observations and wetting properties of the substrates: MO on UV-PDMS exhibited the lowest CA (7°), which resulted in the most favorable infusion capability as also observed by confocal microscopy observations and swelling ratio measurements.

Next, the ability of the biolubricants to confer SLIPS-like characteristics was evaluated by measuring the contact angle hysteresis ($\Delta\theta$) using static water CA measurements at a specific TB angle, which characterizes the resistance to droplet mobility. It is generally considered that an efficient SLIPS coating should exhibit a low contact angle hysteresis ($\Delta\theta < 2.5^\circ$) in order to quickly (within 0.1–1 s) restore liquid repellency after physical damage, consistent

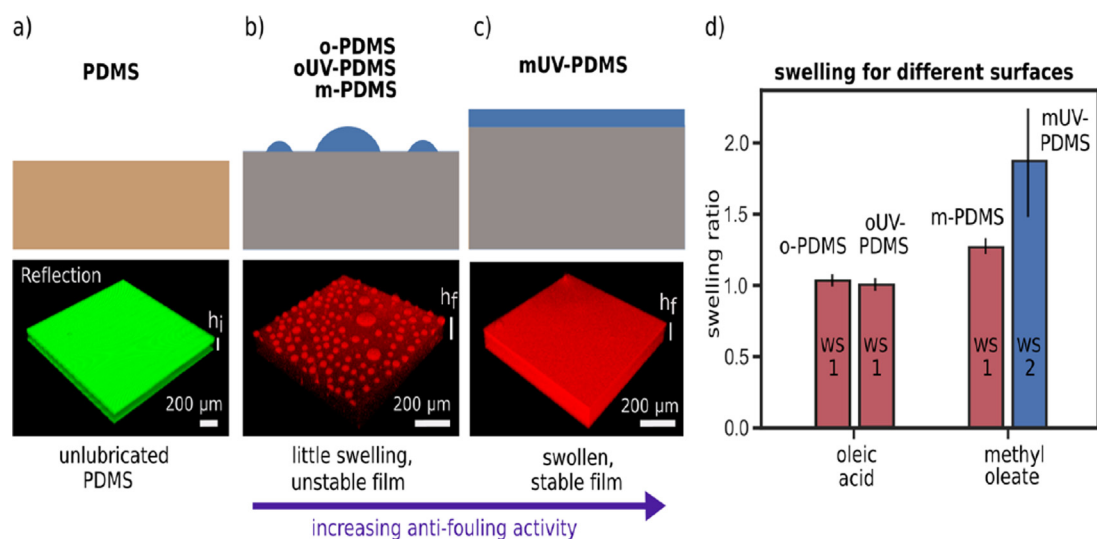


Fig. 1. Schematic and representative fluorescence confocal micrographs of PDMS surfaces infused with various biolubricants. (a) Unlubricated PDMS gel. (b) PDMS gel infused with oleic acid (o-PDMS) or methyl oleate (m-PDMS), or UV-treated PDMS infused with oleic acid (oUV-PDMS). (c) PDMS gel infused with methyl oleate (mUV-PDMS). Confocal microscopy images in the bottom panels (either in reflection and fluorescence mode) show the wetting states, as well as the initial gel thickness h_i and the final thicknesses after infusion h_f . (d) Swelling ratios after infusion with either oleic acid or methyl oleate h_f/h_i , with and without surface UV treatment. Error bars are the standard deviations for 5–6 independent measurements. In panel c), the infused layer remained homogenous for more than one month with no visible de-wetting detected.

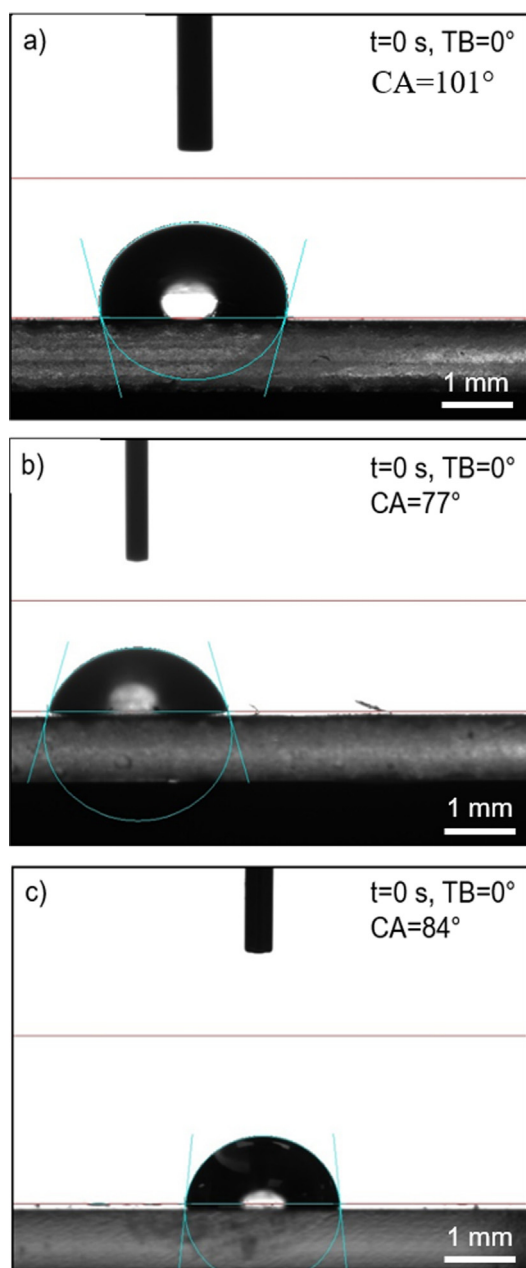


Fig. 2. Contact angle (CA) of static water droplets on different surfaces. (a) UV-PDMS. (b) oUV-PDMS. (c) mUV-PDMS. t = time; TB = Tilting Base angle.

Table 1

Surface and interfacial energies. The surface energies were measured experimentally using the OWRK method, from which the interfacial energies were calculated using Eq. (1).

Sample	DC mN/m	PC mN/m	SE mN/m	IE mN/m
PDMS	19.1	0.08	19.1	–
iPDMS	18.0	0.04	18.0	0.13
oPDMS	17.9	14.3	32.2	14.4
mPDMS	22.9	7.8	30.8	8.1
UV-PDMS	23.8	0.66	24.5	–
oUV-PDMS	19.8	10.8	30.6	11.6
mUV-PDMS	23.8	4.3	28.1	4.9

DC: Dispersive component, PC: Polar component, SE: Surface energy, IE: Interfacial energy.

with a nearly defect-free slippery surface [12]. The higher interfacial free energy (11.63 mN/m) of oUV-PDMS surface resulted in a slow water droplet movement on the surface, likely due to the inhomogeneous layer OA lubricant on the surface (Fig. 1b, Supplementary Movie S2), and to a relatively high $\Delta\theta$ value of 3.7° (Fig. 3a). Therefore, oUV-PDMS had moderate slippery characteristics, which remained well below that of PDMS infused with synthetic oils. In contrast, mUV-PDMS exhibited a 57% lower interfacial energy (4.97 mN/m) than the oUV-PDMS surface (Table 1), leading to a more uniform slippery material (Supplementary Movie S3) with a lower contact angle hysteresis $\Delta\theta$ ($\approx 2^\circ$) (Fig. 3b), therefore confirming its stronger potential as a slippery surface.

3.2. FTIR spectra of OA and MO infused samples

To further confirm infusion of PDMS with the two biolubricants, we carried out ATR-FTIR measurements. UV treated PDMS (Fig. 4a) exhibited the same IR peaks as untreated PDMS at $789\text{--}796\text{ cm}^{-1}$ ($-\text{CH}_3$ rocking and $\text{Si}-\text{C}$ stretching in $\text{Si}-\text{CH}_3$); $1020\text{--}1074\text{ cm}^{-1}$ ($\text{Si}-\text{O}-\text{Si}$ stretching); 1260 cm^{-1} ($-\text{CH}_3$ deformation in $\text{Si}-\text{CH}_3$); and 2965 cm^{-1} (asymmetric $-\text{CH}_3$ stretching in $\text{Si}-\text{CH}_3$) as described previously [30]. The ATR-FTIR spectra of pure OA (Fig. 4b) contained peaks at 2852 and 2922 cm^{-1} attributed to the asymmetric and symmetric CH_2 stretches, respectively, whereas the band at 1409 cm^{-1} corresponds to the $-\text{CH}_3$ umbrella mode. The most intense characteristics peak at wavenumber 1710 cm^{-1} corresponds to the stretching mode of $-\text{C}=\text{O}$. Finally, a broad characteristic band at 3300 cm^{-1} revealed the presence of vibrational stretching assigned to the hydroxyl group of carboxylic acid in OA [31] (Fig. 4b). OA-infused SLIPS sample (Fig. 4c) contained all the relevant peaks associated with the non-infused UV-PDMS and pure OA controls, except for the $-\text{OH}$ broad peak at 3300 cm^{-1} , whose weak initial intensity was masked by the strong absolute intensity of PDMS vibrations.

The IR spectra of MO (Fig. 4d) showed all characteristics peaks associated with OA, except for the band at 3300 cm^{-1} belonging to the hydroxyl group of carboxylic acid ($-\text{COOH}$), which is replaced by an ester group in MO. Accordingly, a band at 1172 cm^{-1} attributed to $-\text{C}-\text{O}-\text{C}-$ moiety was detected. The MO-infused slippery sample (mUV-PDMS) contained all the relevant peaks associated with the non-infused UV-PDMS and pure MO controls (Fig. 4e). The enhanced infusion of MO compared to OA was evidenced by the stronger peak intensities for the MO-infused sample. For example, taking the intensity ratio of the most intense peak of UV-PDMS ($\text{Si}-\text{CH}_3$ at 790 cm^{-1}) to that of OA or MO ($-\text{C}=\text{O}$ of oleic acid at 1710 cm^{-1}) as a proxy for infusion efficiency, we obtained 1:13 and 1:8 for OA- and MO- infused samples, respectively.

3.3. Mussel multichoice attachment assay

To assess whether the biolubricant-infused surfaces were capable of deterring marine biofouling in the lab, we conducted multiple choice checkerboard assays using the Asian green mussel *P. viridis* as a model macrofouling organism. The assay consisted in placing mussels on a board containing multiple slides of different surfaces and observing the settlement behavior of mussels over a 2 weeks period. *P. viridis* is an abundant marine mussel along the shores of the tropical Indo-Pacific that has extremely aggressive fouling characteristics on solid substrates [32]. Three infused SLIPS samples (iPDMS, oUV-PDMS and mUV-PDMS) were considered for the mussel settlement study, followed by adhesion strength measurements. As a control substrate, UV-PDMS surface was selected. Three checkerboards were immersed in aquaria filled with artificial seawater (Supplementary Fig. S5), onto which mussels were placed

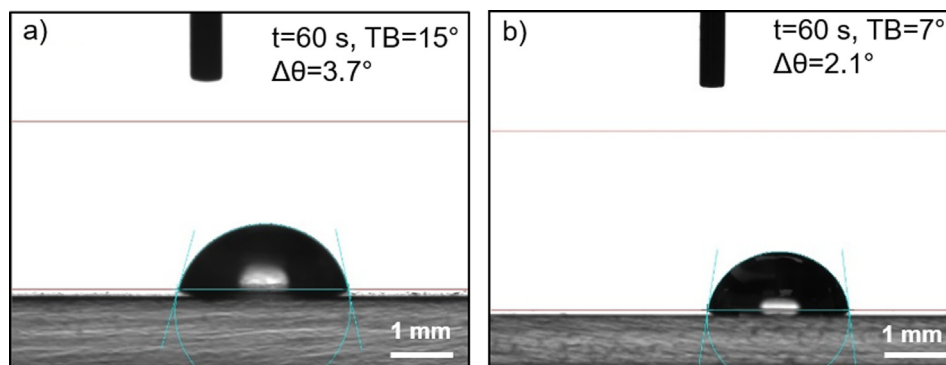


Fig. 3. Optical micrographs demonstrating the mobility of a water droplet on different SLIPS surfaces. (a) oUV-PDMS and (b) mUV-PDMS. t = time; TB = Tilting Base angle, $\Delta\theta$: Contact Angle hysteresis.

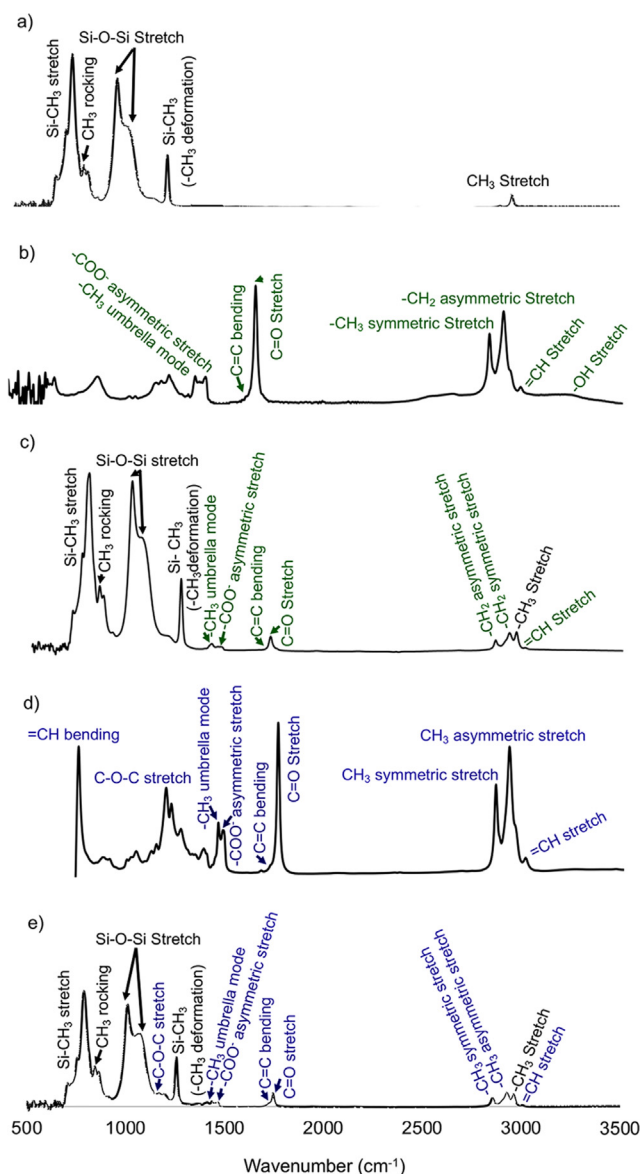


Fig. 4. FTIR spectra of (a) UV-PDMS; (b) oleic acid, (c) oUV-PDMS; (d) methyl oleate, and (e) mUV-PDMS.

with regular spacing. Subsequently, mussels dynamically explore the substrates and choose the most suitable surface to secrete their adhesive threads (Supplementary Fig. S6), using their feet to probe

the surfaces. After two weeks, mussels mostly aggregated into two distinct clusters on the three checkerboards (Fig. 5a). The screening assay showed that the number of mussel adhesive plaques was the smallest on iPDMS (only 26 plaques deposited in total), followed by mUV-PDMS surfaces that contained 76 plaques in total (Fig. 5b). In contrast, the largest number of plaques was found on the UV-PDMS surfaces (total of 340 plaques), while an intermediate number was found on oUV-PDMS surfaces (258 plaques). All data are summarized in Supplementary Table S1. In terms of mean adhesive plaques per slide (n_{ap}) we obtained the following values (Fig. 6a-b): iPDMS, $n_{ap} = 1.4 \pm 0.4$; mUV-PDMS, $n_{ap} = 4.2 \pm 0.9$; oUV-PDMS, $n_{ap} = 13.2 \pm 2.1$; UV-PDMS $n_{ap} = 18.9 \pm 2.5$.

3.4. Plaques adhesive strength

The attachment of *P. viridis* to solid surfaces is enabled by the secretion of byssal threads, which at their distal proximity form the adhesive plaques made by *P. viridis* foot proteins Pvfps [13,33,34]. The ease of mussel detachment (in the instances where mussels did secrete byssal threads), and hence efficacy of a foul-release coating, is directly governed by the plaque/substrate adhesive strength (σ_{ad}): the lower σ_{ad} the better the coating performance. We determined σ_{ad} of *P. viridis* mussel plaques (Fig. 6c) using a custom-made microtensile testing. On iPDMS, we previously measured [13] very weak adhesion values of $\sigma_{ad} = 3.4 \pm 2.0$ kPa. Under identical conditions, we found σ_{ad} of mUV-PDMS to be higher than iPDMS, but still low, namely below 10 kPa ($\sigma_{ad} = 7.1 \pm 0.8$ kPa). In contrast, oUV-PDMS surfaces exhibited significantly higher adhesion strength ($\sigma_{ad} = 16.5 \pm 1.2$ kPa), whereas for non-infused surface (UV-PDMS) the values were much higher and similar to the previous measurements ($\sigma_{ad} = 31.7 \pm 2.4$ kPa). The highest adhesive strength was obtained for the bare glass surface ($\sigma_{ad} = 82.6 \pm 6.7$ kPa). These adhesive strength data further validate the anti-fouling properties deduced from the checkerboard settlement assay. Namely, mUV-PDMS is the most promising surface of all coatings produced in this study, with a performance only slightly inferior than iPDMS. mUV-PDMS is not only more efficient to deter mussel's attachment, but release of the few attached plaques can be achieved with relative ease.

3.5. Nanoscale contact forces

Beyond interfacial energy considerations, biofouling is also controlled by the mechanical properties of the coating, namely surface stiffness and capillary forces arisen due to the lubricant layer. To measure the surface stiffness, we carried out depth-sensing nanocontact mechanic measurements using a spherical indenter geometry with a 10 μm nominal radius, and the modulus was

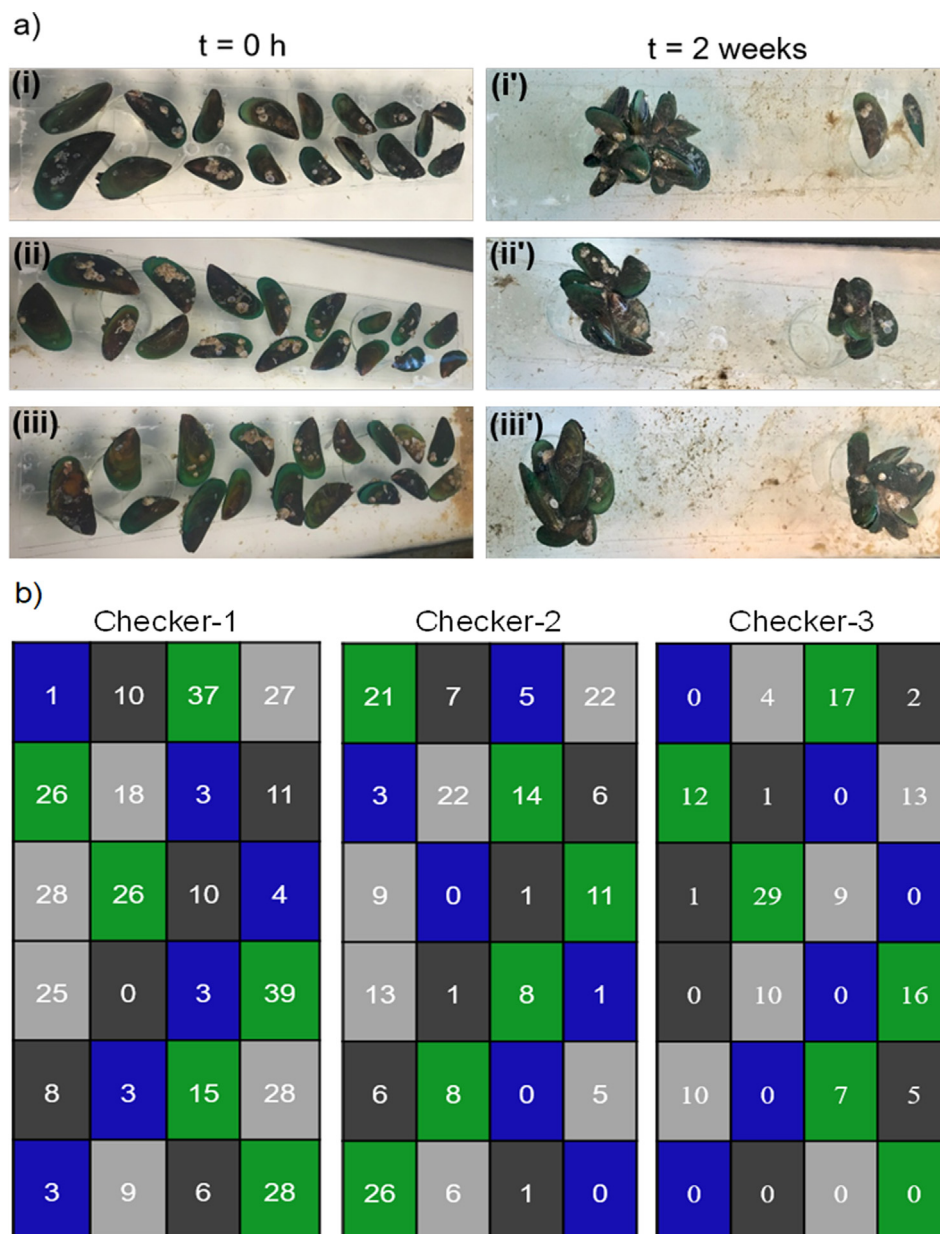


Fig. 5. Checkerboard assay and plaque secretion of *P. viridis*. (a) Multiple choice assay illustrating the randomized checkerboard arrangement of the various surfaces onto which mussels were uniformly placed (i-iii) at time zero (left) and allowed to move and settle (i'-iii') for 2 weeks (right). Three checkerboards were prepared, each consisting of 6 slides of 4 surface types, with slides randomly distributed on each checker. (b) Quantitative results of checkerboard multiple choice assay. The number on each slide corresponds to the number of adhesive plaques counted after 2 weeks of testing. Surface color codes: UV-PDMS (green), iPDMS (blue), oUV-PDMS (light grey) and mUV-PDMS (dark grey).

obtained by fitting the load-displacement curves with the Hertzian equation [25] up to $ca.$ 3 μm contact depth. Characteristic load-displacement curves are shown in Fig. 7a. Upon infusion with the biolubricants, the modulus of UV-treated PDMS decreased from $1.9 \pm 0.2 \text{ MPa}$ to 1.3 and 0.74 MPa for oUV-PDMS and mUV-PDMS, respectively. The lower modulus of the MO-infused surface can be attributed to its higher swelling ratio, since the elastic modulus of swollen networks is well-known to decrease with swelling [35]. A second critical property of lubricant-infused surfaces can be inferred from depth-sensing nanoindentation measurements, namely adhesive and capillary forces forming at the solid/liquid interface. Indeed, we previously posited that the formation of a capillary bridge at the interface can “deceive” the mechano-sensing ability of the mussel foot, thereby deterring the animals

from secreting byssal threads [13]. During a loading/unloading cycle, the contact mechanics signature of a capillary bridge are forces in the adhesive regime during retraction from the surface over a significantly large distance. In a depth-sensing contact mechanics experiment, this translates into negative forces as well as negative displacements (zero displacement being the surface, and positive displacements corresponding to the tip in compressive contact with the surface), as shown in Fig. 7b-d. Whereas the capillary adhesive forces between OA-infused and MO-infused PDMS surfaces were nearly equivalent, we measured much larger capillary bridges for mUV-PDMS ($H = 12.5 \mu\text{m}$, Fig. 7d) compared to oUV-PDMS ($H' = 2 \mu\text{m}$, Fig. 7c), in line with the more efficient infusion of MO in the UV-treated PDMS substrate. We note that the capillary height for mUV-PDMS is very similar to that

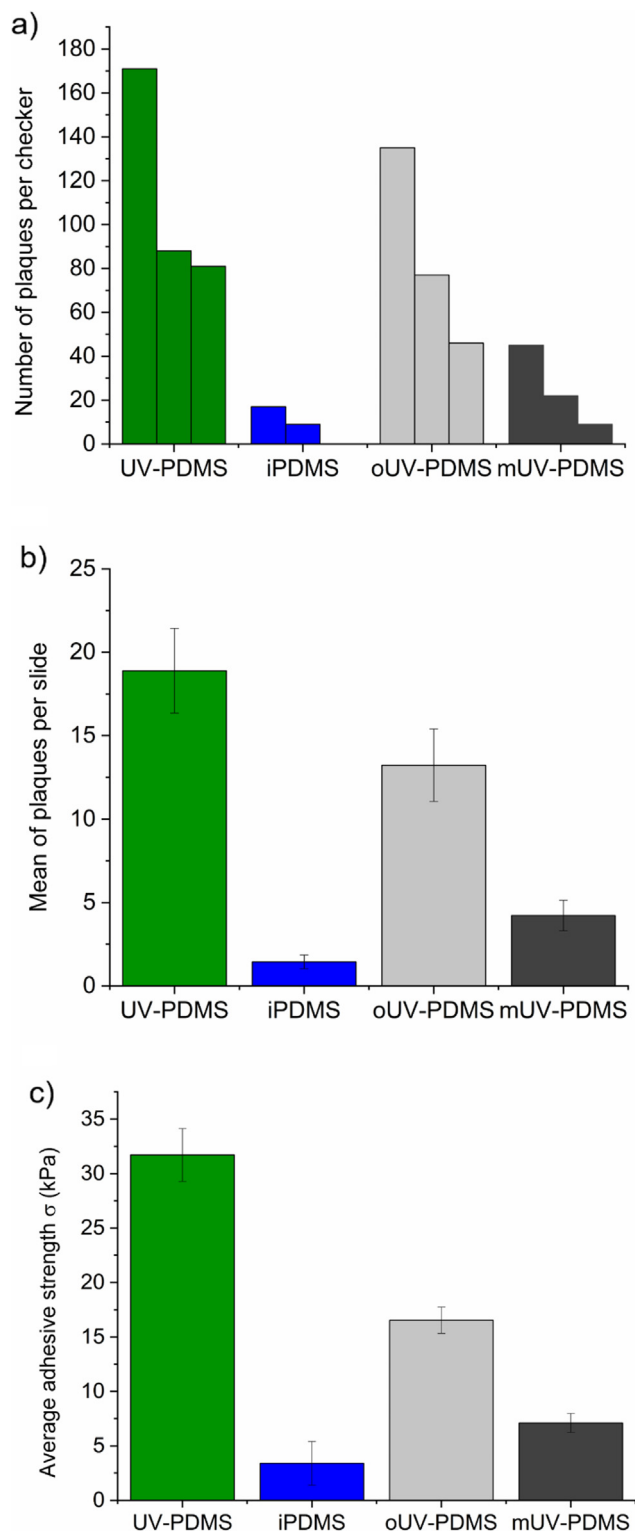


Fig. 6. Number of (a) plaque per checker and (b) plaques per slide on the different surfaces, and (c) average adhesive strength σ (kPa) on the different surfaces. For (c), the error bars are standard error (SE) values obtained from five replicates.

previously measured for iPDMS ($H \sim 13 \mu\text{m}$), again corroborating the comparable infusion ability of MO with the synthetic fluorinated lubricant.

3.6. Cytotoxicity against fish gill cell

To determine the potential toxicity of the surfaces, we used a fish gill cell line as a predictor of fish toxicity [26] and subjected these cells to two cytotoxicity assays: the alamarBlue cell viability reagent (used as an indicator of living cells' health) and the LDH cytotoxicity kit (which detects lactate dehydrogenase released in the cell culture supernatant during cytoplasmic membrane damage). Fish gill cell viability indicated that the biolubricant-infused coatings were non-cytotoxic (Fig. 8a). In particular, the LDH assay showed that the plasma membranes of the cells were not disrupted by the presence of OA and MO lubricants (Fig. 8b), indicating that any leachate from the coatings would not be toxic (Fig. 8c).

4. Discussion

The anti-fouling activity of fatty acids was first reported more than 25 years ago [36], where a mixture of fatty acids isolated from a marine sponge was shown to deter the attachment of byssal threads from the blue mussel *Mytilus edulis*. Unsaturated fatty acids showed better antifouling ability than saturated ones [36]. However, the observations were limited to just three mussels left overnight on surfaces incubated with the fatty acid mixture and the mechanism of repellency was not established. OA, an unsaturated fatty acid, is safe for living cells. It is the main fatty component of olive oil and presents useful healthy characteristics. For example, it can boost immunomodulation or help in the treatment and prevention of different types of disorders such as cardiovascular or autoimmune diseases, metabolic disturbances, skin injury and cancer [37]. The esters of fatty acids have also been registered with the Food and Drug Administration (FDA) for use as a direct food additive (21 CFR 172.225) and in animal feed (21 CFR 573.640), strongly suggesting that MO is safe to marine life. In the present study, we further verified the biocompatibility of OA and MO by assessing the cell viability of a fish gill cell line with a cytotoxicity assay developed to predict fish toxicity [26] (Fig. 8). No cell death was observed for the leachates of both OA and MO-infused surfaces, suggesting that these coatings are safe to the marine life.

OA was initially identified as a biolubricant in our study due to its structural similarity to PDMS. Both components have surface-active (low surface energy), closely packed methyl groups that are arranged along a more flexible backbone. This common feature should enhance the chemical affinity and facilitate spreading and retention of the lubricant through Van der Waals and capillary forces. oUV-PDMS tended to phase separate on the PDMS surface, resulting in an inhomogeneous lubricant layer (Supplementary Movie S4). To improve infusion, we selected MO because we hypothesized that the absence of the hydrophilic carboxylate group would result in a lower surface energy, which would be closer to that of PDMS, a central requirement for the SLIPS concept [12]. mUV-PDMS indeed displayed better chemical affinity as confirmed by the physical smoothness of the lubricant layer at the liquid-liquid interface (Fig. 1 and Supplementary Movie S5). The superior infusion ability of MO within the PDMS network was corroborated by the higher swelling ratio (Fig. 1d), which in turn translated into a significant smaller elastic modulus for mUV-PDMS compared to oUV-PDMS (Fig. 7a). All together, these properties eventually resulted in a very low contact angle hysteresis ($\Delta\theta < 2.5^\circ$) at a lower sliding angle (7°) compared to OA infused SLIPS (Fig. 3). The contact angle hysteresis and the sliding angle directly characterize the resistance to mobility; the low values

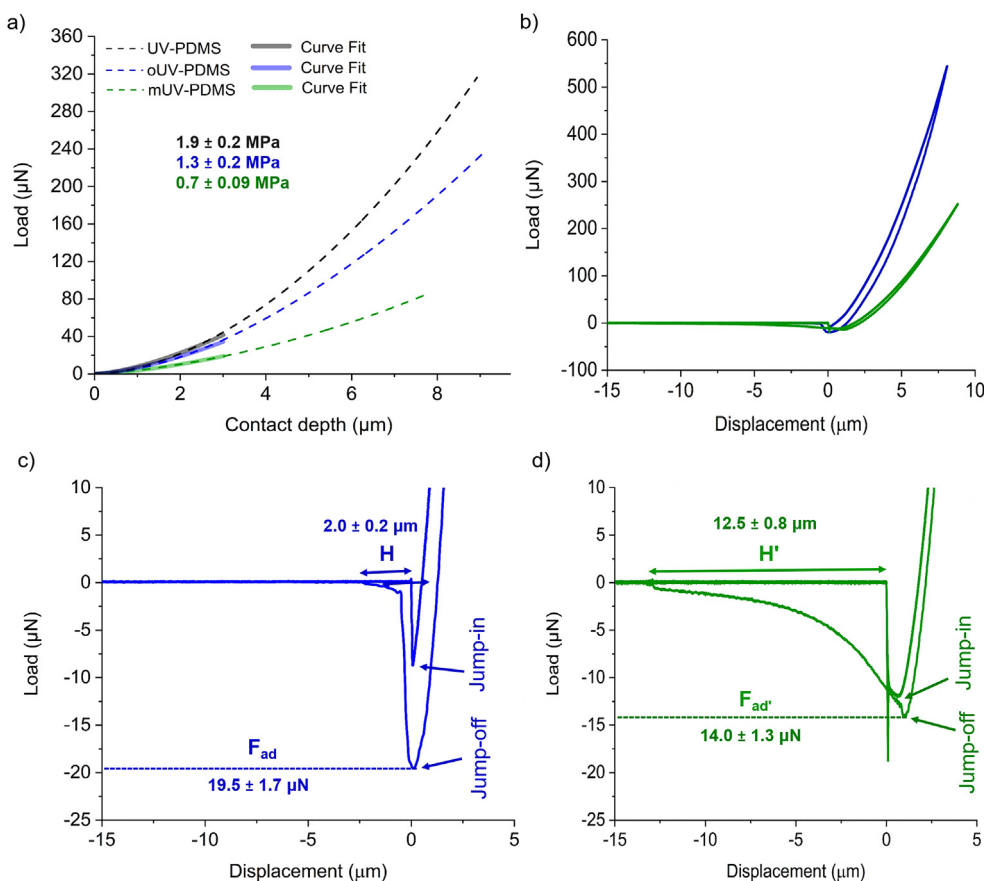


Fig. 7. Nanoscale contact mechanics of UV-PDMS, oUV-PDMS and mUV-PDMS infused surfaces. (a) Load-displacement curves obtained with a cono-spherical tip ($10\ \mu\text{m}$ radius). The elastic modulus was inferred from the loading portion of a loading/unloading cycle using the Hertzian solution. (b) Characteristic load-displacement curves obtained with a cono-spherical tip with $50\ \mu\text{m}$ nominal radius used in order to detect capillary forces. (c-d) Enlargement of the capillary adhesive force (F_{ad} and $F_{ad'}$) regimes for oUV-PDMS (c) and mUV-PDMS (d) slippery surfaces. Jump-in and jump-off instabilities upon approach and retraction, respectively, are attributed to capillary bridges of the lubricant. H and H' are the capillary bridge lengths. Standard deviation (SD) values were calculated from seven replicates.

therefore confirmed a lack of pinning, consistent with a nearly defect-free surface [38]. The UV irradiation treatment increased the surface energy of PDMS, minimized the substrate/lubricant interfacial tension, and hence improved the wetting of the surface by the lubricant (Supplementary Fig. 4). While we have opted to use UV irradiation in this study, in principle any surface treatment that increases the surface energy – including chemical modifications and plasma treatment – will improve the wetting of the lubricant. We note that although hydrophilic groups created by UV irradiation are typically short-lived in air, MO remained entrapped within the UV-treated PDMS substrate for more than a month. Previous work indicates that the hydrophilic groups are more stable in the presence of polar molecules, such as water and ethanol (Ref. [29]). It is therefore possible that the ester groups in the infused MO stabilized the hydrophilic moieties on the PDMS surfaces. The detailed physical mechanisms behind the outstanding antifouling activity of silicone oil-infused PDMS was previously attributed to the ultra-low interfacial energy of the substrate/lubricant interface [13], which provides a two-pronged strategy to prevent mussel fouling. First, a low interfacial energy favors the permanent infusion of the lubricant, resulting in a thicker lubricant layer entrapped at the interface, which in turn provides “shielding” against detection of the surface from the mechano-sensing mussel foot organ. Second, a lower interfacial adhesion leads to a smaller thermodynamic work of adhesion w_a , which directly correlates with the macroscopic adhesion strength of mussel threads σ_{ad} . Therefore, by decreasing w_a the macroscopic adhesion is reduced,

and thus ease of mussel detachment is enhanced. According to these two criteria, one would expect a direct correlation between the interfacial energy of the (bio)lubricant/substrate interface and the antifouling performance indexes, here defined as mean number of deposited adhesive plaques (n_{ap}) and plaque adhesion strength (σ_{ad}). Plotting these two parameters against the interfacial energy (Fig. 9) for the UV-treated surfaces prepared in this study as well as iPDMS, we observe a clear correlation with a high R value for both indexes, confirming that a low interfacial energy between the substrate and the lubricant is a central criterion in order to inhibit mussel biofouling. In particular, minimizing the interfacial energy leads to a more uniform lubricant layer, as observed here when comparing mUV-PDMS (uniform lubricant layer) with oUV-PDMS (uneven distribution of lubricant micro-droplets in the substrate). It is likely that a more homogenous lubricant layer minimizes the probability that mussels can detect the underlying substrate when they are probing the surface.

Although the mUV-PDMS surface was not as performant as iPDMS in preventing mussel biofouling (Fig. 7d), its more hydrophilic characteristics ($CA = 84^\circ$ vs. 110° for iPDMS) may have other benefits beyond its eco-friendly features. For example, it may enhance the antifouling against other adsorbed proteins since most proteins that contaminate solid surfaces are hydrophobic in nature. The less hydrophobic properties of mUV-PDMS may result in a hydration layer forming on these coated infused surfaces, which may help repulsing non-specific proteins and act as an additional antifouling mechanism.

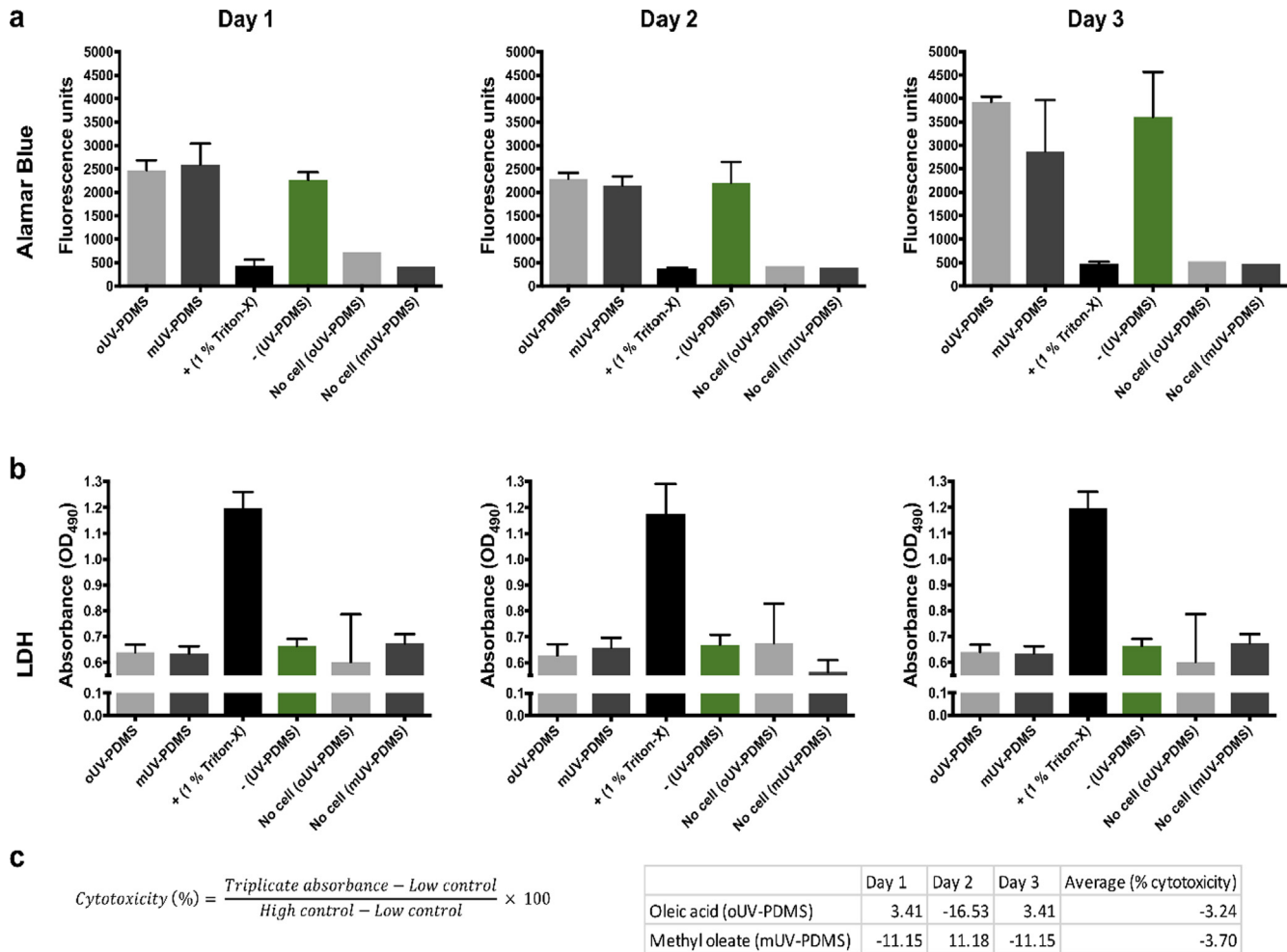


Fig. 8. Cytotoxicity study of the SLIPS surfaces using fish gill cells. Rainbow trout gill cells RTgill-W1 were exposed to cell culture media containing leachate from oUV-PDMS (light grey) and mUV-PDMS (dark grey) for up to 3 days (72 h). The controls were cell culture medium supplemented with 1% Triton-X (black) as a positive control (+) for cell death, and non-infused UV-PDMS (green) as a no killing control (–) to show that the porous surface itself is non-toxic. There were also no cell controls representing background signals from the test solutions. (a) Cell viability measured using the alamarBlue assay determines cellular metabolic activity. (b) LDH assay measures the plasma membrane damage. (c) The cytotoxicity of the leachate was calculated from the LDH assay values. All values obtained were a mean of triplicates except for the no cell control of the alamarBlue assay, and the error bars are the standard deviations of the values obtained.

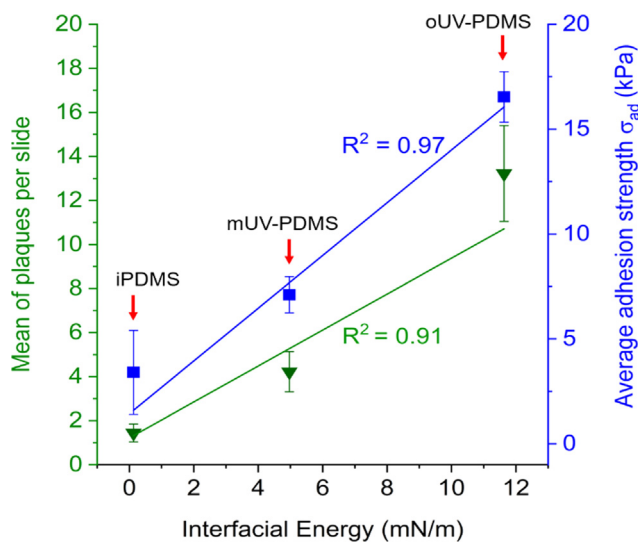


Fig. 9. Number of adhesive plaques per slide (n_{ap}) and mean plaque adhesive strength (σ_a) vs. the interfacial energy. Adhesion strength data from iPDMS are replotted from Amini et al. [13].

5. Conclusion

Lubricant infused surfaces have gathered considerable interest in recent years. Whereas the majority of these studies have focused on substrate engineering and applications of these coatings in a wide range of fields, the development of specific eco-friendly lubricants for the infused phase has been less explored [18]. In the present study we hypothesized that the fatty acids OA and MO could represent alternatives to synthetic oils to infuse PDMS substrates and combat marine biofouling with harmless chemicals. Modifying the surface hydrophilicity as well as tuning the biolubricant chemistry led to enhanced wetting of the biolubricant by minimizing the substrate/biolubricant interfacial energy. The resulting biolubricants-infused coatings were able to deter the attachment of marine mussels (one of the most pervasive macrofouling organisms) and to minimize the adhesion strength of their byssal threads. It was also verified that the coatings did not exhibit cytotoxicity against a fish gill cell line, used as a proxy to establish fish acute toxicity. In addition, these coatings may also exploit the natural ability of saturated oils to inhibit bacterial biofilm, which we are currently exploring. This study demonstrates that low cost, environmentally friendly, and readily available fatty acid precursors hold attractive potential as alternatives to synthetic

halogenated lubricants currently used in the vast majority of slippery surfaces. Achieving an anti-fouling performance only slightly weaker than with synthetic lubricants while developing an eco-friendly alternative is considered an acceptable compromise, especially in the light of growing demands to mitigate the environmental impact associated with the accumulation of man-made chemicals in the sea. While the present proof-of-concept study establishes the potential of biolubricant-based SLIPS to mitigate marine biofouling in laboratory settings, multi-months field testing will be required to establish the fouling-resistance efficacy for practical applications.

CRedit authorship contribution statement

Snehasish Basu: Conceptualization, Methodology, Validation, Investigation, Visualization, Data curation, Formal analysis, Writing - original draft. **Bui My Hanh:** Validation, Investigation, Data curation. **J.Q. Isaiah Chua:** Methodology, Software. **Dan Daniel:** Methodology, Validation, Visualization. **Muhammad Hafiz Ismail:** Methodology, Validation, Visualization. **Manon Marchioro:** Validation. **Shahrouz Amini:** Methodology. Software. **Scott A. Rice:** Conceptualization, Supervision, Funding acquisition. **Ali Miserez:** Conceptualization, Supervision, Writing - original draft, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

We declare no conflict of interest with the submission of our article.

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Appendix A. Supplementary material

Schematic of the confocal microscope setup in the reflection and fluorescence modes (Figure S1); contact angles on untreated PDMS surface (Figure S2); confocal microscopy images of oPDMS and mPDMS (Figure S3); oil infusion (OA and MO) on plain PDMS and UV-PDMS at room temperature (Figure S4); checkboard before the multiple choice assay (Figure S5); adhesives plaques on infused surfaces after 2 weeks of the multiple choice assay (Figure S6); total number of adhesive plaques on surfaces (Table S1); water droplet on PDMS surface (Movie S1); water droplet on oUV-PDMS (Movie S2); water droplet on mUV-PDMS surface (Movie S3); biolubricant micro-droplets for oUV-PDMS (Movie S4); uniform homogeneous biolubricant layer for mUV-PDMS (Movie S5).

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2020.02.049>.

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